

Cloud LearnX

Mathematics

Transformations — Practice Worksheet

Student: Amy

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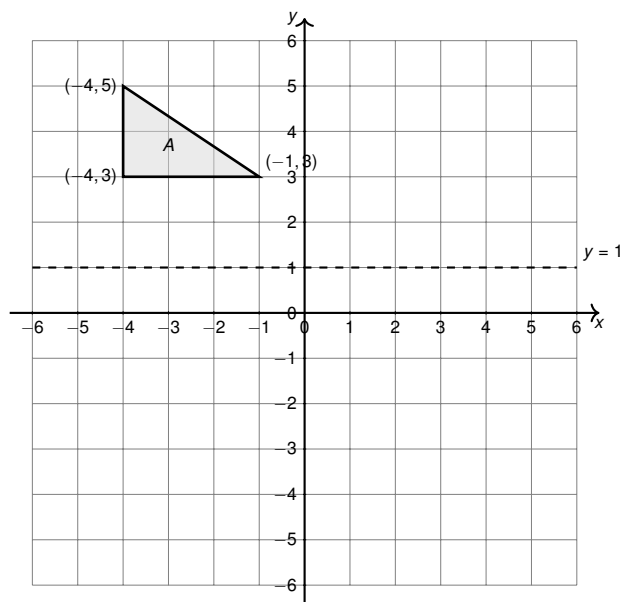
Topic: Transformations

Instructions:

- Answer ALL questions.
- Show ALL working — method marks are awarded for correct steps.
- Write answers in the spaces provided.

Answer ALL ALL questions.
Write your answers in the spaces provided.
You must write down all the stages in your working.

- 1** Triangle A has vertices at $(-4, 3)$, $(-4, 5)$ and $(-1, 3)$.



Reflect triangle A in the line $y = 1$.

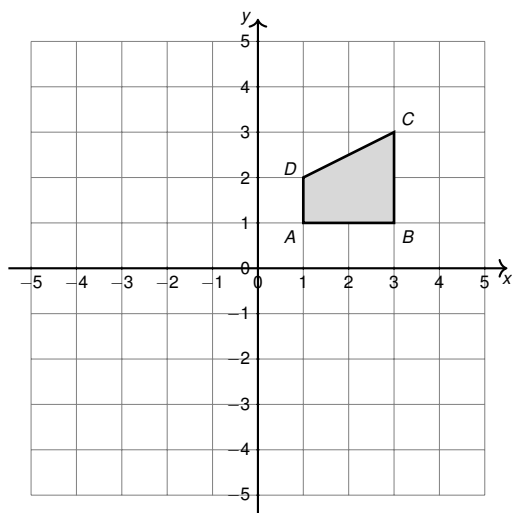
Find the coordinates of the image of each vertex of triangle A .

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(2)

(Total for Question 1 is 2 marks)

2 Quadrilateral $ABCD$ is shown on the grid below.



Rotate quadrilateral $ABCD$ by 90° clockwise about the origin O .

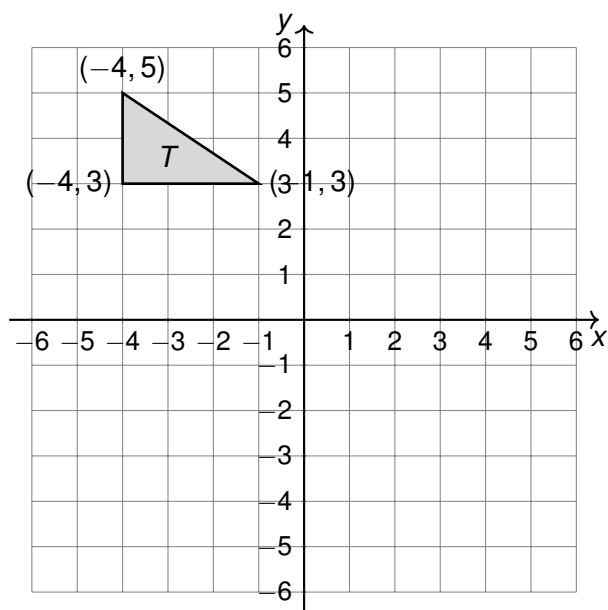
Write down the coordinates of the image of each vertex.

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(3)

(Total for Question 2 is 3 marks)

3 Triangle T is drawn on the grid below.



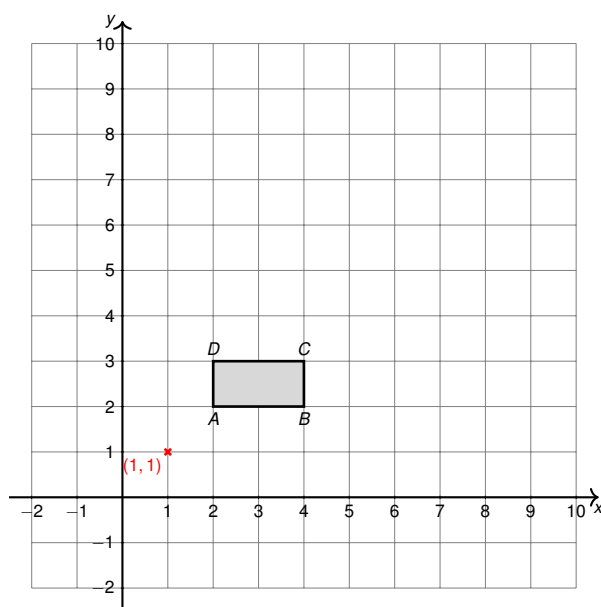
Triangle T is translated by the vector $\begin{pmatrix} 3 \\ -2 \end{pmatrix}$ to give triangle T' .

Find the coordinates of the vertices of triangle T' .

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(2)

(Total for Question 3 is 2 marks)

- 4 Rectangle $ABCD$ has vertices $A(2, 2)$, $B(4, 2)$, $C(4, 3)$ and $D(2, 3)$. The point $(1, 1)$ is marked with a cross as the centre of enlargement.



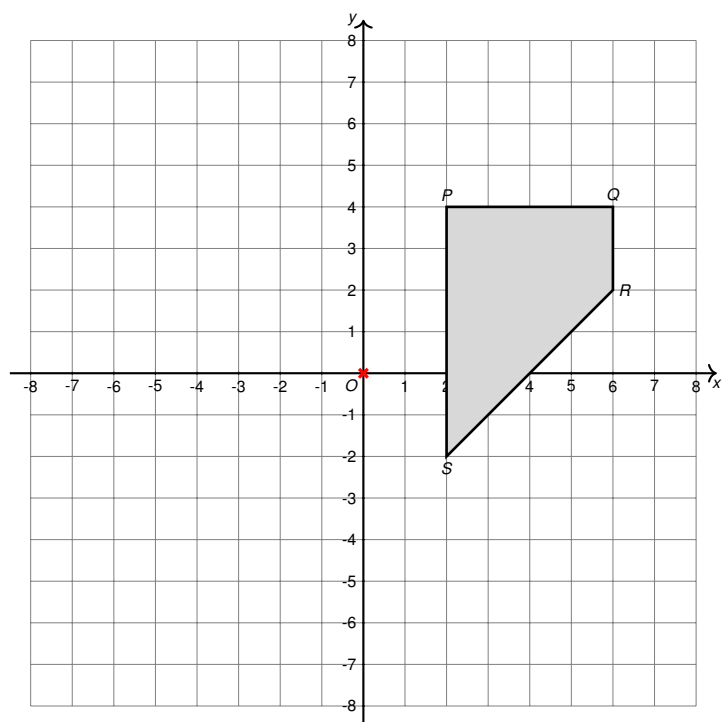
Enlarge rectangle $ABCD$ by scale factor 2, centre $(1, 1)$.

Work out the coordinates of the vertices of the image, A' , B' , C' and D' .

.....
(3)

(Total for Question 4 is 3 marks)

5 Quadrilateral $PQRS$ is shown on the grid below.



Quadrilateral $PQRS$ has vertices $P(2, 4)$, $Q(6, 4)$, $R(6, 2)$ and $S(2, -2)$.

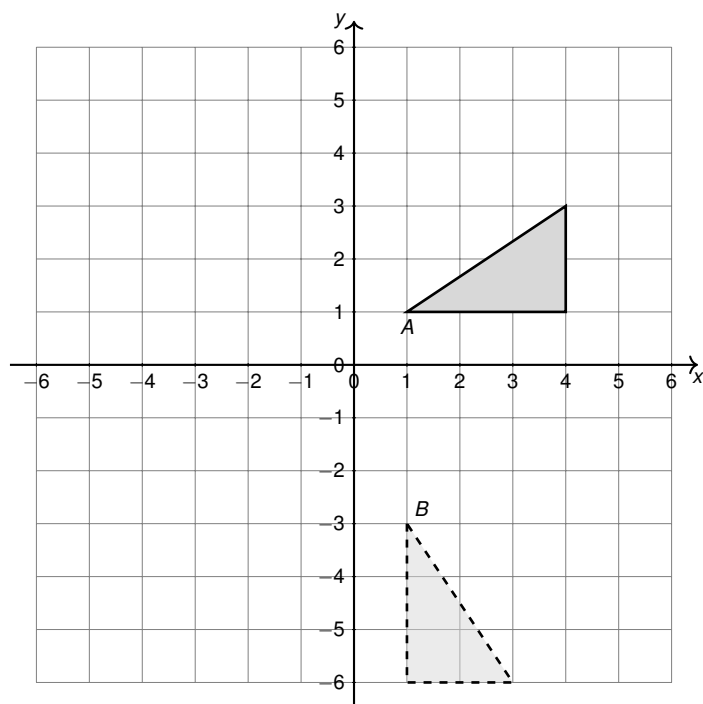
Enlarge $PQRS$ by scale factor $\frac{1}{2}$, centre $(0, 0)$.

Write down the coordinates of the image of each vertex.

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(3)

(Total for Question 5 is 3 marks)

6 Shape A and shape B are drawn on the grid below.



(a) Find the type of transformation that maps shape A onto shape B.

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(1)

(b) Fully describe the single transformation that maps shape A onto shape B.

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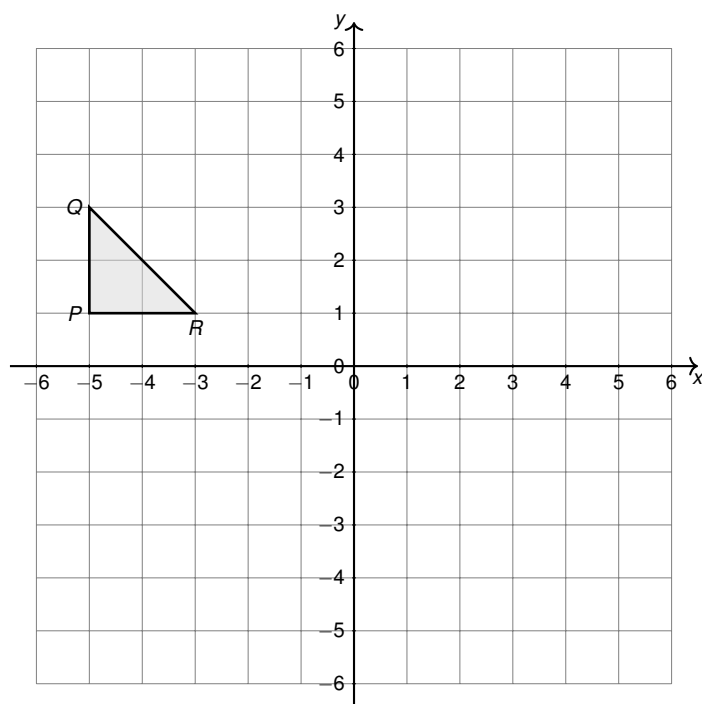
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(3)

(Total for Question 6 is 4 marks)

- 7 A tracking station plots the ground position of a satellite as triangle PQR on a coordinate grid, shown below.



The vertices of triangle PQR are $P(-5, 1)$, $Q(-5, 3)$ and $R(-3, 1)$.

- (a) Triangle PQR represents an orbital shift and is translated by the vector $\begin{pmatrix} 2 \\ 3 \end{pmatrix}$ to give triangle $P'Q'R'$.
Work out the coordinates of P' , Q' and R' .

.....

(2)

- (b) Triangle $P'Q'R'$ is then reflected in the line $x = 0$, representing a mirrored sensor reading, to give triangle $P''Q''R''$.
Calculate the coordinates of P'' , Q'' and R'' .

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(2)

(Total for Question 7 is 4 marks)

TOTAL FOR PAPER IS ?? MARKS